STREAMTIME

Sovereign Streaming

Running an enterprise grade private real-time data estate.

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Executive Summary

Streamtime is a **management plane** for private real-time data estates.

A set of meta-layer, workload-aware operational APIs for multi-tenant streaming, stream processing and real-time analytics. The controller turns workload intent into placement, lifecycle, policy and operational actions where throughput, latency and failure behavior matter.

IMPACT

Security posture

Private networking, restricted egress and air-gap modes reduce externally reachable management paths while preserving governed operations.

IMPACT

Operational flexibility

Define workload SKUs, choose topology, placement and storage, bring your own enterprise peripherals, and optimize against measured substrate performance.

IMPACT

Cost leverage

Retain open infrastructure economics while productizing fleet operations that would otherwise require a specialist platform team.

Business impact

Adopting Streamtime compounds security posture and operational flexibility with cost savings. The operating model is designed to deliver the leverage of a specialist real-time platform team with only a fraction of the dedicated headcount.

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The argument

The deck moves from the operating constraints of private real-time data estates to the technical primitives Streamtime provides to run them as a coherent platform.

CONTEXT

Streaming is now part of the default data stack

Real-time data is increasingly the live context layer for applications, decisions and agents. That moves streaming beyond a niche reserved for only latency-sensitive use cases.

BFSI

payments, fraud, risk, customer events

Healthcare

clinical events, devices, operational telemetry

Manufacturing

machine telemetry, quality, predictive maintenance

Government

case events, cyber telemetry, citizen services

Digital business

CDC, product events, recommendations, automation

Agentic systems

fresh context, tool events, state changes, feedback loops

Why this matters now

The platform has to carry performance, locality and control at the same time. Private real-time estates are therefore a first-class operating model, especially in regulated and infrastructure-heavy industries.

Source: kafka.apache.org — Kafka is used by thousands of companies and more than 80% of the Fortune 100.

CONTEXT

There is still a choice problem

The current landscape forces enterprises to trade control, risk and operating cost against each other.

OPERATING MODEL

Managed PaaS

Fastest to consume

Provider controls service topology

Usage + capacity premiums

OPERATING MODEL

Enterprise distribution

Software + support contract

Customer owns 24x7 operations

License + platform team + services

OPERATING MODEL

OSS self-managed

Maximum component freedom

Customer owns incidents and upgrades

Infra + platform team + services risk

STREAMTIME

Streamtime

Workload-aware management plane

24x7 fully managed operations

Open infra economics + global Data/SRE

The gap Streamtime targets

Private estates usually choose between software plus reactive support, or a managed service that owns the infrastructure boundary. Streamtime keeps the estate under customer control while adding a 24x7 fully managed operating layer backed by global Data/SRE teams.

What sovereignty means

Infrastructure	**Where it runs**	on-prem • BYOC • edge • air-gap
Operations	**Who controls lifecycle**	provision • upgrade • recover • troubleshoot
Dependencies	**What can be replaced**	identity • PKI • storage • observability • models

Sovereignty is the compound of all three.

The practical outcome

A sovereign real-time platform keeps data-plane placement, operational authority and dependency choices inside the enterprise boundary, while still offering a consistent management contract across environments.

ECONOMICS

Economics: open engines are cheap. Operating real-time estates is not.

Kafka, Flink and ClickHouse are efficient open systems. Cost accumulates around the estate: compute and storage, network transfer, managed-service margin, commercial licensing, professional services and the people required to run lifecycle and incidents across multiple engines.

Infrastructure	compute, storage, network, load balancers and data movement
Managed-service margin	capacity minimums, usage margins and service premiums across engines
Licensing	commercial streaming, processing and analytics distributions and support
Support + services	case-driven support plus architecture, implementation and optimization services
Operations + risk	upgrades, rebalances, checkpoints, incidents, capacity, compliance and 24x7 ownership

Published pricing and support models expose the same pattern: infrastructure is only one line item. Managed services meter streaming, processing and analytics on separate usage dimensions; enterprise distributions add software, support and professional services while the customer still owns production responsibility.

The Streamtime objective

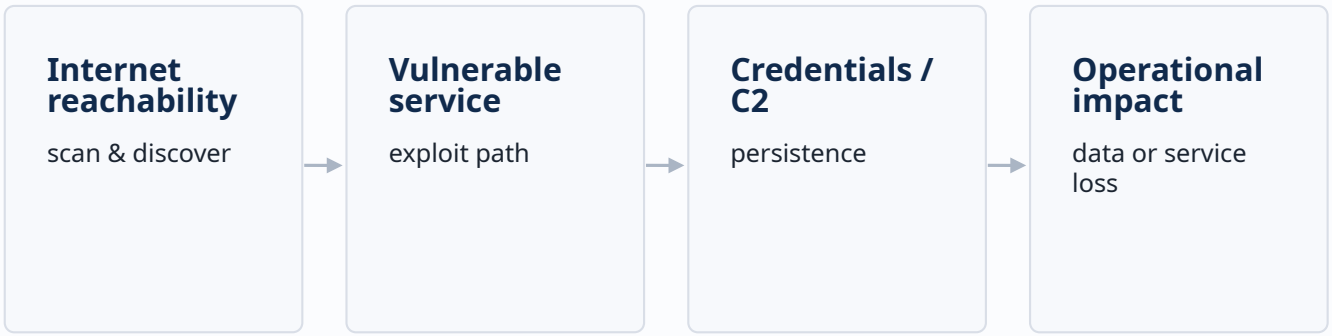
Preserve open infrastructure economics across streaming, stream processing and real-time analytics while productizing the operational layer. As estates add workloads, clusters and throughput, duplicated operations, support costs and per-service premiums compound faster than raw infrastructure cost.

Sources: aws.amazon.com/msk/pricing • confluent.io/services • docs.confluent.io/cloud/current/support.html

SECURITY

Privacy, zero-days and air-gapped operation

Zero-day exposure is a timing problem. Before patches and reliable detections exist, reachable services can be discovered and exploited at machine speed.



Air-gapped mode removes the remote path.

It does not remove supply-chain, insider, removable-media or controlled-transfer risk. It changes the attack surface from internet-scale reachability to explicitly governed ingress and egress.

Why it remains effective against malicious agents

Agents can accelerate reconnaissance and exploitation, but they still need a reachable path. Air-gapping removes internet-scale scanning, remote exploit delivery and command-and-control from the operating model.

Source: csrc.nist.gov/glossary/term/Zero_Day_Attack

CONTROL

Operational freedom: define workload SKUs, optimize, benchmark

Workload SKUs	Define supported SKUs across streaming, stream processing and analytics: engine/runtime profile, topology, storage, connectors and SLOs.
Optimize topology	Choose placement, zones, storage, network and failure domains for each workload and the local substrate.
Benchmark reality	Measure throughput, latency, state/checkpoint behavior and headroom on the actual substrate before setting capacity expectations.
Operate from evidence	Keep saturation curves and steady-state baselines with each workload so scaling, recovery and tuning decisions have context.

Freedom to optimize

Sovereignty gives the platform freedom to optimize each workload against its actual substrate. Streamtime ties workload SKU, topology and capacity decisions to measured performance instead of assuming that provider labels imply equivalent behavior.

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One management plane for real-time data

DESIGN NOTE

One fleet contract, three workload types. Streaming, stream processing and real-time analytics keep their runtime semantics while sharing placement, tenancy, policy, observability, lifecycle and API surfaces.

WORKLOAD TYPE

Streaming

Apache Kafka • Kafka Connect • MirrorMaker
durable events • ingestion • fan-out

WORKLOAD TYPE

Stream processing

Apache Flink • Kafka Streams
stateful compute • windows • enrichment

WORKLOAD TYPE

Real-time analytics

ClickHouse • Iceberg-compatible data
interactive queries • serving • exploration

A common operating model

The engines stay specialized. The platform contract around them is shared: fleet identity, placement, tenancy, lifecycle, policy, observability, cost attribution and stable APIs.

Streamtime management plane

DESIGN NOTE

Control is out of band; execution stays local. Desired state, policy and operation history live in the management plane. Fleet agents execute inside the customer boundary over Headscale-coordinated WireGuard with relay fallback when direct paths fail.

OUT OF BAND

Management plane

fleet API
desired state
policy + operation history

NETWORK

Private overlay

Headscale coordination
WireGuard direct paths
DERP fallback

CUSTOMER BOUNDARY

Fleet agent

local execution
health + inventory
engine adapters

Customer data plane

Kafka • Flink • ClickHouse • Kubernetes • storage • network • identity • observability

Interfaces

UI | REST API | SDK | CLI | IaC | MCP server

The architectural boundary

Streamtime owns management state and operational intent. Workload data and execution remain inside the customer estate. The same model can extend from connected BYOC to restricted egress and fully disconnected deployments.

Source: tailscale.com/docs/reference/derp-servers — direct paths preferred; encrypted relay is a fallback when direct connectivity is unavailable.

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Multi-tenancy is a core architectural premise

DESIGN NOTE

Tenancy is explicit at two layers: compute-pool isolation and logical workload isolation. Shared, isolated and dedicated modes govern compute placement; virtual clusters provide logical multi-tenancy within a workload.

FAILURE MODES

Without a tenancy model

Cluster sprawl duplicates brokers, storage, certificates, upgrades and monitoring. Mega clusters reduce count but increase noisy-neighbor risk, blast radius, coordination cost and attribution ambiguity.

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With a tenancy model

Workloads choose shared, isolated or dedicated compute pools. Virtual clusters add logical multi-tenancy where the engine supports it, while the fleet keeps one management plane, policy model and operational history.

Shared

Same physical fleet and shared-services compute pool.

Isolated

Same physical fleet and shared services; isolated workload compute pool.

Dedicated

Dedicated compute pool; one tenant pinned.

Virtual clusters

Logical multi-tenancy within the workload layer.

Fleet-level controls placement • quotas + capacity • RBAC + approvals • chargeback • blast radius • maintenance

The architectural point

Tenancy belongs in the architecture before the estate grows. Compute-pool isolation defines resource and blast-radius boundaries; virtual clusters provide logical tenancy without forcing every tenant into a separate physical cluster.

A platform contract around the operator

DESIGN NOTE

Runtime operators sit behind a stable platform contract. Streamtime adds declarative configuration, tenancy, bring-your-own peripherals, lifecycle state, operational context and stable APIs around engine-specific reconciliation.

Declarative intent

throughput • retention • availability • topology

Enterprise peripherals

OIDC • PKI • DNS • storage • observability • licenses

Durable operations

create • scale • upgrade • recover • delete

Operational context

desired vs actual • metrics • history • ownership • policy

Why this matters

Private estates need configuration freedom without turning every workload into a snowflake. The platform contract keeps the user-facing model stable while enterprise dependencies remain replaceable underneath.

AUTONOMOUS OPERATIONS

Streamtime Operational Harness

DESIGN NOTE

Operational reasoning starts from measured steady state. The harness combines fleet and workload context with baselines, then investigates and acts through bounded platform APIs, workload-specific Admin Skills and the Streamtime MCP server.

Fleet infrastructure	nodes • storage • network • placement • saturation
Streaming	brokers • partitions • ISR • consumer lag • quotas • connectors
Stream processing	jobs • operators • checkpoints • watermarks • backpressure • state
Real-time analytics / OLAP	replicas • shards • merges • ingest lag • query latency • storage
Cross-stack telemetry	metrics • logs • traces • events • topology • change history

Steady-state baseline + operational context	normal throughput • latency • state/query profile • recent changes • incidents • policy • ownership

Observe → Compare → Reason → Act → Verify

Streamtime Incident Response: automatic investigations

Automatic investigations correlate fleet, streaming, processing and analytics evidence against steady-state baselines, build and rank hypotheses, determine blast radius, propose or execute bounded remediation, verify recovery and preserve the audit trail.



Thank you

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